The edge exponent of the uniform doubling condition and an interpolation conjecture

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Abstract

In a sequence of companion papers we determined the critical scaling of the uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] for i.i.d. samples in two regimes: $\Theta((\log N/N)^{2/(n+1)})$ for pure slabs over smooth convex bodies, and $\Theta((\log N/N)^{1/n})$ in the presence of conical truncations. This note locates the first intermediate exponent. For the unit ball truncated by a *single transversal half-space* — so that the boundary of the truncation region has an $(n-2)$-dimensional edge but no lower-dimensional strata — we prove that with probability $1-o(1)$ the doubling inequality fails at every grid scale below $c\,(\log N/N)^{2/(n+2)}$, $n \geq 3$. The obstruction is the most isolated sample point of the *edge caps*, whose volumes scale as $h^{(n+2)/2}$, intermediate between the smooth cap exponent $\frac{n+1}{2}$ and the conical exponent n. A second-moment argument over $\asymp (N/\log N)^{(n-2)/(n+2)}$ congruent caps along the edge upgrades the failure probability to $1-o(1)$, with the logarithm. We conjecture the general interpolation law: for truncation families whose regions have all boundary strata of dimension at least j and transversal intersections, the critical exponent is $\frac{2}{2n-j}$ — recovering $\frac{2}{n+1}$ for smooth boundaries ($j=n-1$), $\frac{2}{n+2}$ for edges ($j=n-2$), and $\frac{1}{n}$ for conical points ($j=0$) — and we explain what a matching upper bound requires.

1 Setting and result

Let $K = B^n \subset \mathbb{R}^n$ be the closed unit ball, $n \geq 3$, μ the uniform probability measure on K, and $T = \{X_1, \ldots, X_N\}$ i.i.d. samples with empirical measure $\bar{\mu}_N$. Truncated slabs $S_w = \{x : |\langle x - s, u\rangle| \leq \frac{w}{2}\} \cap H$ and the uniform doubling condition (UDC) with constants $(\kappa, \delta, \sigma, \beta, \ell)$ are as in [1, 2]. The critical scaling of the smallest admissible scaling factor ℓ is known in two regimes [3, 4, 5]: exactly $\Theta((\log N/N)^{2/(n+1)})$ for pure slabs ($k=0$) over bodies with a rolling ball, and exactly $\Theta((\log N/N)^{1/n})$ for thick conical truncation geometry. Here we consider the geometrically intermediate case: a single half-space truncation

$$H := \{x : \ x_1 \leq 0\}, \qquad G := K \cap H,$$

whose boundary consists of two smooth $(n-1)$-dimensional pieces (a spherical cap and a flat disk) meeting transversally (at dihedral angle $\pi/2$) along the $(n-2)$-dimensional edge

$$E \ := \ \partial K \cap \partial H \ = \ \{x : \ |x| = 1, \ x_1 = 0\}.$$

The region G is uniformly thick (it contains the ball $B(-\frac{1}{2}e_1, \frac{1}{4})$), and the upper bound of [2] applies: the UDC holds with probability $1-\frac{1}{N}$ at scaling $\ell_N = O((\log N/N)^{1/n})$. We prove that the true critical scale for this single truncation is governed by a new exponent.

Theorem 1.1. *Let $n \geq 3$ and fix $\kappa > 1$, $\sigma > 0$, $\beta > 0$, $\delta \geq \frac{3}{N}$. There are constants $\varepsilon_2(n,\kappa) > 0$ and $c(n,\kappa) > 0$ such that, with probability $1 - o(1)$ as $N \to \infty$, the doubling inequality fails for some slab truncated by the single truncation set $\{(e_1, 0)\}$ at every grid scale below $c\,(\log N/N)^{2/(n+2)}$. Consequently, with probability $1-o(1)$, the UDC for this truncation requires*

$$\ell \;\geq\; c\Big(\frac{\log N}{N}\Big)^{2/(n+2)}.$$

Since $\frac{2}{n+2} > \frac{1}{n}$ would fail and indeed $\frac{2}{n+2} < \frac{2}{n+1}$, the bound interpolates strictly between the two known regimes: a single transversal truncation of a smooth body already destroys the smooth exponent $\frac{2}{n+1}$, but does not produce the conical exponent $\frac{1}{n}$ — for that, by [3], genuinely conical (lowest-dimensional) boundary strata are needed. We record the natural guess suggested by the three data points.

Conjecture 1.2 (Interpolation law)**.** *For i.i.d. samples with density bounded above and below on a convex body, and truncation families of bounded size whose nonempty regions are uniformly thick and have all boundary strata transversal and of dimension at least j, the critical scaling of the uniform doubling condition is*

$$\ell_N \;\asymp\; \Big(\frac{\log N}{N}\Big)^{2/(2n-j)},$$

where for $j = n-1$ (smooth, rolling-ball boundary) this is the proven $\frac{2}{n+1}$ [4], for $j = n-2$ (transversal edges) it is the exponent $\frac{2}{n+2}$ of Theorem 1.1, and for $j = 0$ (conical points) it is the proven $\frac{1}{n}$ [3, 5].

The heuristic behind the law is uniform: the critical resolution is the isolation depth of the most isolated extremal sample point, and a cap of depth h at a j-dimensional transversal stratum has volume $\asymp h^{(2n-j)/2}$ — each of the $n-1-j$ transverse flat directions contributes a factor h, the normal direction contributes h, and each of the j curved stratum directions contributes $\sqrt{h}$. Theorem 1.1 verifies the count for $j = n-2$: $h^2 \cdot h^{(n-2)/2} = h^{(n+2)/2}$.

2 Edge caps

Fix the angle $\varphi = \frac{\pi}{4}$ and, for a point $p \in E$, the direction

$$u_p \;:=\; \tfrac{1}{\sqrt{2}}\,e_1 + \tfrac{1}{\sqrt{2}}\,p.$$

Note $\langle x, u_p\rangle$ is maximized over G at $x = p$, with maximum $\frac{1}{\sqrt{2}}$: for $x \in G$, $\langle x, u_p\rangle = \frac{1}{\sqrt{2}}\big(x_1 + \langle x, p\rangle\big) \leq \frac{1}{\sqrt{2}}\big(0 + |x|\,|p|\big) \leq \frac{1}{\sqrt{2}}$, with equality iff $x_1 = 0$ and $x = p$. Define the *edge cap*

$$C_p(h) \;:=\; \Big\{x \in G :\; \langle x, u_p\rangle \geq \tfrac{1}{\sqrt{2}} - h\Big\}.$$

Lemma 2.1 (Edge cap volume)**.** *There are constants $0 < c_8(n) \leq c_9(n)$ such that for all $p \in E$ and $0 < h \leq h_0(n)$,*

$$c_8\,h^{\frac{n+2}{2}} \;\leq\; \mu\big(C_p(h)\big) \;\leq\; c_9\,h^{\frac{n+2}{2}}.$$

Proof. By rotational symmetry about the e_1-axis we may take $p = e_2$. Write $x = (x_1, x_2, x'')$ with $x'' \in \mathbb{R}^{n-2}$, and $h' := \sqrt{2}\,h$. The cap is

$$C_p(h) = \Big\{ x : \ |x| \le 1,\ x_1 \le 0,\ x_2 - |x_1| \ge 1 - h' \Big\}$$

(using $x_1 \le 0$ to write $x_1 = -|x_1|$). On $|x| \le 1$ we have $x_2 \le \sqrt{1 - x_1^2 - |x''|^2} \le 1 - \frac{x_1^2 + |x''|^2}{2}$, and $x_2 \ge 1 - h' + |x_1|$ forces

$$|x_1| \ \le\ h', \qquad |x''|^2 \ \le\ 2h', \qquad 1 - h' + |x_1| \ \le\ x_2 \ \le\ \sqrt{1 - x_1^2 - |x''|^2}. \tag{1}$$

Upper bound: the x_2-fibre has length at most $\big(1 - \frac{x_1^2 + |x''|^2}{2}\big) - (1 - h' + |x_1|) \le h'$, so $\mathrm{vol}(C_p(h)) \le h' \cdot (2h') \cdot \omega_{n-2} (2h')^{\frac{n-2}{2}} \le C(n) h^{\frac{n+2}{2}}$, where the three factors bound the x_2-, x_1- and x''-ranges. *Lower bound:* restrict to $|x_1| \le \frac{h'}{8}$, $|x''|^2 \le \frac{h'}{2}$; then, for $h' \le h_0$ small, using $\sqrt{1-t} \ge 1 - t$ for $t \in [0,1]$,

$$\sqrt{1 - x_1^2 - |x''|^2} - (1 - h' + |x_1|) \ \ge\ h' - |x_1| - x_1^2 - |x''|^2 \ \ge\ h' - \tfrac{h'}{8} - \tfrac{h'}{64} - \tfrac{h'}{2} \ \ge\ \tfrac{h'}{4},$$

so the fibre length is at least $\frac{h'}{4}$ on a set of (x_1, x'')-measure $\frac{h'}{4} \cdot \omega_{n-2} (\frac{h'}{2})^{\frac{n-2}{2}}$, giving $\mathrm{vol}(C_p(h)) \ge c(n) h^{\frac{n+2}{2}}$. Dividing by $\mathrm{vol}(B^n)$ gives the bounds for μ. □

Lemma 2.2 (Footprint and disjointness)**.** *There is $c_{10}(n)$ such that $C_p(h) \subseteq \{x : \ |x - p| \le c_{10}\sqrt{h}\}$ for $h \le h_0$. Consequently, if $p_1, \dots, p_M \in E$ are pairwise at distance greater than $2c_{10}\sqrt{H}$, the caps $C_{p_j}(H)$ are pairwise disjoint, and on the $(n-2)$-dimensional sphere E one may choose*

$$M \ \ge\ c_{11}(n)\, H^{-\frac{n-2}{2}} \tag{2}$$

such points.

Proof. From (1) (with $p = e_2$): $|x_1| \le h'$, $|x''| \le \sqrt{2h'}$ and $1 \ge x_2 \ge 1 - h' + |x_1| \ge 1 - h'$, so $|x - e_2|^2 \le h'^2 + 2h' + h'^2 \le 4h'$. The packing bound on the $(n-2)$-dimensional sphere E — of dimension $n - 2 \ge 1$ for $n \ge 3$ — is the standard volumetric estimate. □

Lemma 2.3 (Stall)**.** *Fix $\kappa > 1$, $\sigma > 0$, $\delta \ge \frac{3}{N}$, $p \in E$, and depths $0 < h \le \frac{H}{4}$, $H \le h_0$. Suppose exactly one sample point lies in $C_p(H)$ and that point, X^*, lies in $C_p(h)$. Then for every $w \le H/\kappa$ the slab with direction u_p, center X^*, and truncation $\{(e_1, 0)\}$ satisfies $\bar\mu_N(S_w) = \bar\mu_N(S_{\kappa w}) = \frac{1}{N} < \min\{\delta, (1+\sigma)\bar\mu_N(S_w)\}$; hence the doubling inequality fails at every scale $w \le H/\kappa$, and the UDC with scaling factor $\ell \le H/\kappa^2$ fails.*

Proof. For $w' \le H$: $S_{w'} \subseteq G \cap \{\langle x, u_p\rangle \ge \langle X^*, u_p\rangle - \frac{w'}{2}\} \subseteq C_p\big(h + \frac{w'}{2}\big) \subseteq C_p(H)$, since $h + \frac{w'}{2} \le \frac{H}{4} + \frac{H}{2} < H$; the rest is as in [4, 5]. □

3 Proof of Theorem 1.1

Proof. Set $q := (4\kappa^2)^{\frac{n+2}{2}} \frac{c_9}{c_8}$, $\varepsilon_2 := \frac{n-2}{2(n+2)\,q}$ and $\tau_N := \varepsilon_2 \log N$. Define h_N by $\mu(C_p(h_N)) = \frac{\tau_N}{N}$ (the cap mass is continuous and strictly increasing in h, and independent of p by symmetry); by Lemma 2.1, $h_N \asymp (\tau_N/N)^{2/(n+2)}$. Set $H_N := 4\kappa^2 h_N$, so by Lemma 2.1, $\mu(C_p(H_N)) \le q\,\frac{\tau_N}{N}$.

Choose $p_1, \dots, p_M \in E$ as in Lemma 2.2 with $M \geq c_{11} H_N^{-\frac{n-2}{2}} \geq c_{12}(N/\tau_N)^{\frac{n-2}{n+2}}$, and let

$$E_j := \Big\{ \#\big(T \cap C_{p_j}(H_N)\big) = 1,\ T \cap C_{p_j}(H_N) \subseteq C_{p_j}(h_N) \Big\}, \qquad X := \sum_{j \leq M} \mathbf{1}_{E_j}.$$

Exactly as in [4]:

$$p := \mathbb{P}(E_j) = N \cdot \frac{\tau_N}{N} \big(1 - \mu(C_{p_j}(H_N))\big)^{N-1} \geq \tau_N\, e^{-2q\tau_N} = \tau_N\, N^{-2q\varepsilon_2} = \tau_N\, N^{-\frac{n-2}{n+2}},$$

so $\mathbb{E}X = Mp \geq c_{12}\, \tau_N^{1-\frac{n-2}{n+2}} \to \infty$; replacing ε_2 by $\varepsilon_2/2$ gives the safer $\mathbb{E}X \geq c_{12} N^{\frac{n-2}{2(n+2)}} \to \infty$. The caps being disjoint, the two-cap correlation ratio is $\exp(O(N\mu(C(H_N))^2)) = \exp(O((\log N)^2/N)) = 1 + o(1)$ as in [4], so $\mathbb{P}(X = 0) \leq \frac{1}{\mathbb{E}X} + o(1) = o(1)$ by Chebyshev. With probability $1 - o(1)$ some E_j occurs, and Lemma 2.3 forces $\ell > H_N/\kappa^2 = 4h_N \geq c(n,\kappa)(\log N/N)^{2/(n+2)}$. □

4 Towards the matching upper bound

Conjecture 1.2 predicts that for the single transversal truncation of Theorem 1.1 the UDC in fact *holds* with high probability at scaling $\ell_N = O((\log N/N)^{2/(n+2)})$. By the proof skeleton of [2, 4], this reduces to a deterministic geometric estimate: a uniform lower bound

$$\mathrm{vol}\Big(G \cap \big\{y :\ |\langle y - x, u\rangle| \leq L\big\}\Big) \ \geq\ c\, L^{\frac{n+2}{2}} \qquad (x \in G,\ u \in S^{n-1},\ 0 < L \leq L_0),$$

the analogue for $G = K \cap H$ of the rolling-ball slab-volume bound of [4]. The anchor–unimodality framework of [4] reduces this to lower bounds on the level-section volumes $A(\tau)$ of G at depth h below the u-extreme, for every direction: the required anchor value is $A \gtrsim h^{n/2}$ at some depth $h \leq L/4$. Three regimes occur: extremes on the smooth spherical part (where the rolling-ball sections give the stronger rate $h^{(n-1)/2}$, but the half-space may cut them), extremes on the flat disk (where the sections contain $(n-1)$-dimensional disk caps of volume $\asymp h^{n/2}$), and extremes on the edge (where the explicit computation of Lemma 2.1 gives sections of volume $\asymp h^{n/2}$). A convex-segment witness from the extreme point to the deepest point of the cut rolling ball shows that the section's penetration depth into H grows at least linearly in h, which controls the cut in the first regime. Assembling these into a single uniform constant — in particular through the transition directions where the extreme point migrates from the spherical part across the edge to the flat disk — requires a careful case analysis that we have not completed, and which we leave, together with the general law of Conjecture 1.2, as the open problem of this note. We emphasize that all three regime computations individually produce the predicted exponent $\frac{n+2}{2} = \frac{2n-j}{2}$ with $j = n-2$, in agreement with the stratum-counting heuristic.

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

References

[1] Y. Shmalo, *Stability of accuracy for the training of DNNs via the uniform doubling condition*, Ann. Math. Artif. Intell. (2024), DOI 10.1007/s10472-023-09919-1.

[2] Y. Shmalo, *The uniform doubling condition holds with high probability for independent samples*, manuscript, 2026.

[3] Y. Shmalo, *The critical scaling of the uniform doubling condition: smooth bodies versus truncations*, manuscript, 2026.

[4] Y. Shmalo, *The logarithm is necessary: exact critical scaling of the uniform doubling condition on the ball*, manuscript, 2026.

[5] Y. Shmalo, *The exact critical scaling of the uniform doubling condition with truncations*, manuscript, 2026.